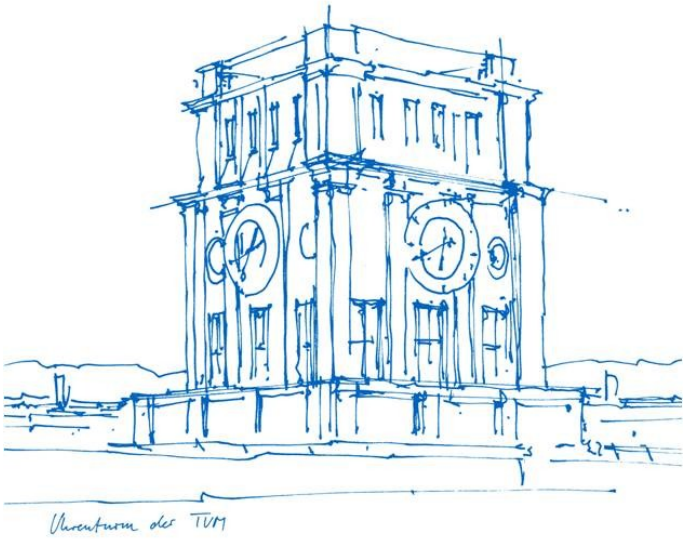
A Case-Grounded Benchmark for Evaluating LLM-Based GeoXAI Agents

Condensed Master's Thesis Defense Presentation

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Evaluation framework

Each question combines one Stage with one Evidence Tier.

What task is assessed?

01 Task and data understanding

02 Data and spatiotemporal analysis

03 Model behavior explanation

04 Reliability and applicability

What evidence is needed?

Sanity

General concepts

Discriminative

Case-specific results

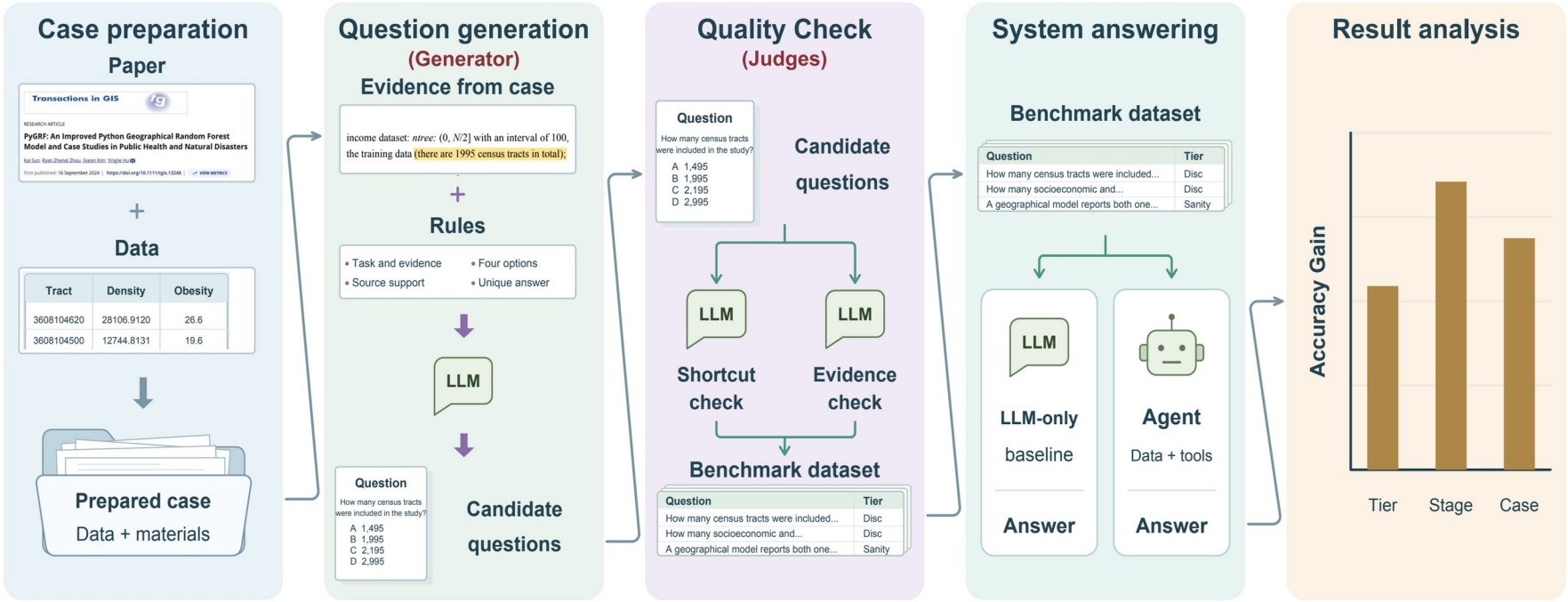
Bespoke

Combining case findings

Stages describe tasks; Tiers describe evidence and analysis requirements.

Automated benchmark workflow

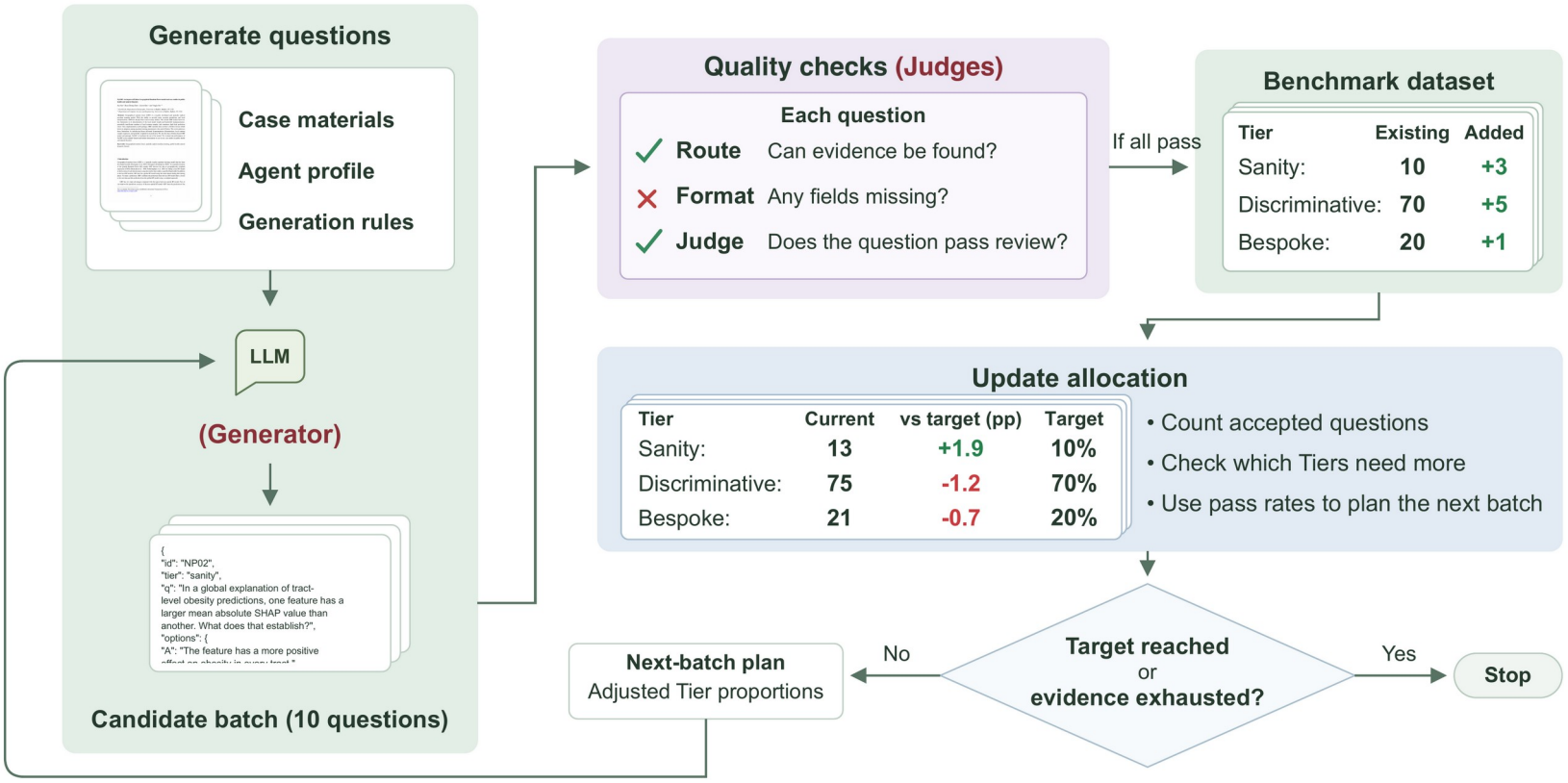
From published studies to reviewed questions and paired evaluations.



Questions keep evidence records for review. The same questions are used to compare GeoXAI-Agent with the corresponding LLM-only baseline.

Step 2: Question generation

Accepted questions and review outcomes guide the next generation batch.



Question record example

A question, its reference answer and the supporting source evidence.

Question record structure

Example: NYC obesity case

Metadata

id	type	tier	stage
NP11	mcq	disc	task_data_understanding

Question and options

q How many census tracts were included in the NYC obesity case-study dataset?

options

- **A** 1,495 census tracts
- **B** 1,995 census tracts
- **C** 2,195 census tracts
- **D** 2,995 census tracts

Reference answer

answer **B**

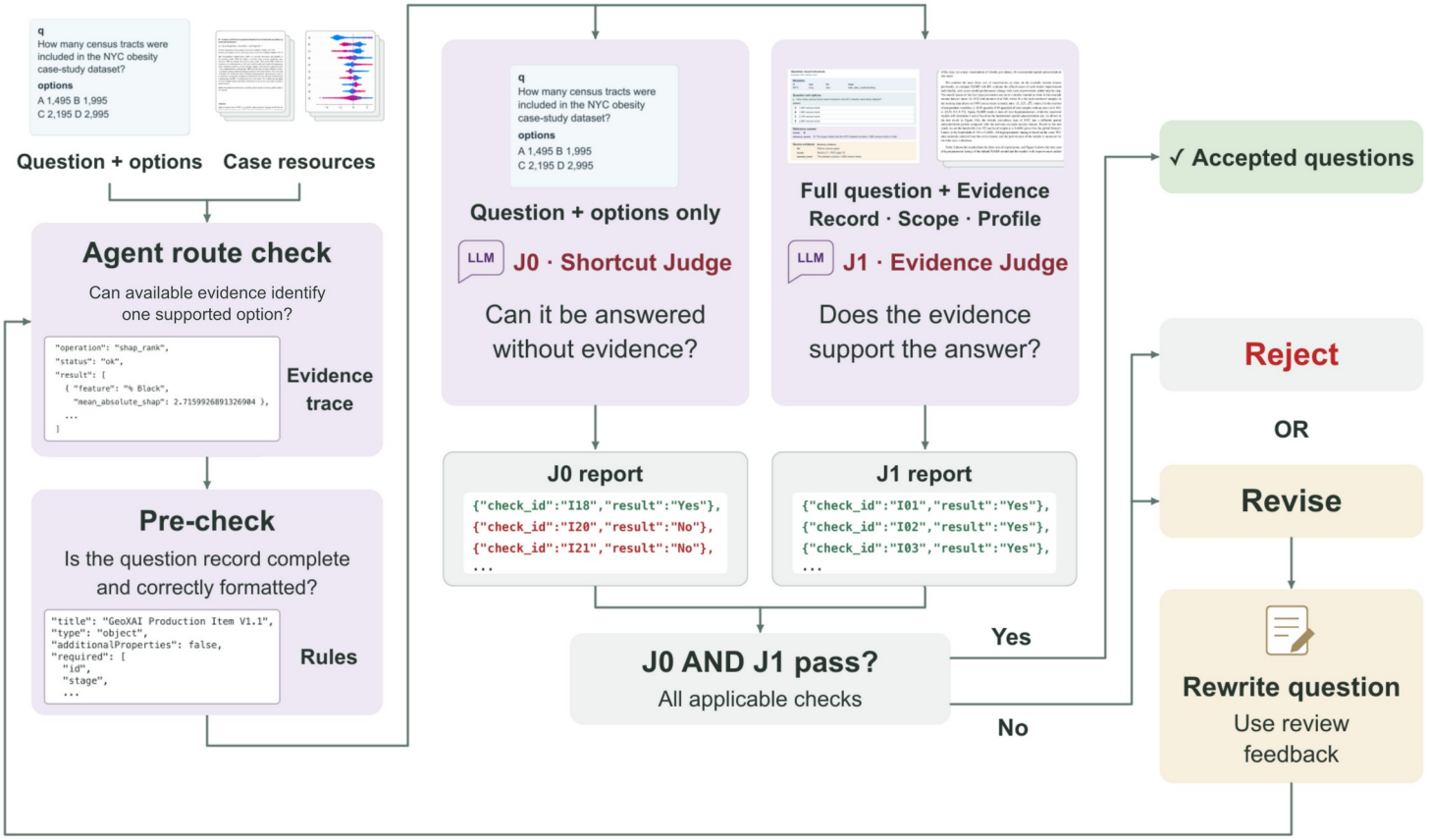
reference_answer B. The paper states that the NYC dataset contains 1,995 census tracts in total.

Source evidence **literature_evidence**

file	Path to source paper
locator	Section 5.1, PDF page 16
reported_result	The dataset contains 1,995 census tracts.

Step 3: Quality check

Route and format checks precede shortcut and evidence review.



A question that failed human review

Human review found an unsupported local-effect claim and an incomplete Bespoke evidence chain.

Question example

Case: Chicago ride-hailing Stage: Model behavior explanation Tier: **Bespoke**

Which synthesis matches education across the paper's **citywide model findings** and the registered **focal-tract** explanation?

Reference answer (D)

Education is **globally leading**, has a **strong positive fitted relationship**, and **pushes the focal prediction downward**

Supporting evidence

Education is **one of the most important features** and has a **strong positive association** with ride-hailing demand.

Reviewer-checked location: Figures 11–12, PDF page 13

Human review

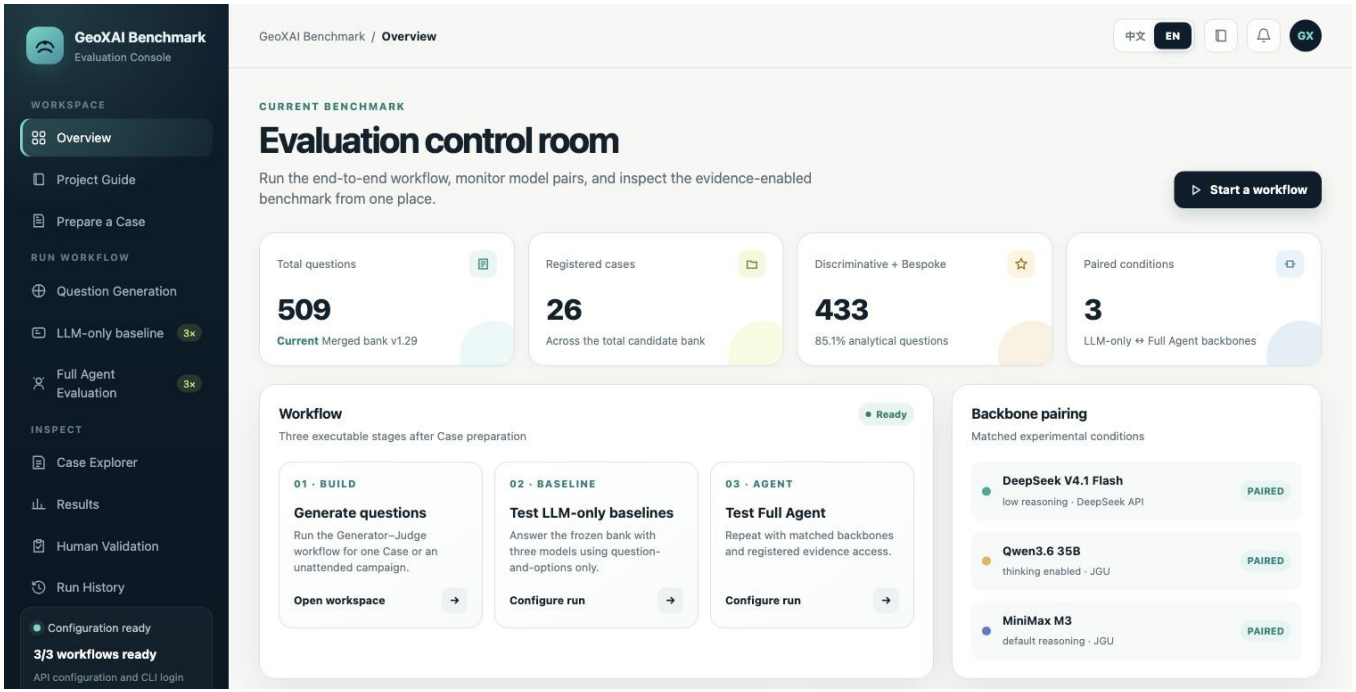
Clarity	✓	
Evidence consistency	✗	The paper **does not support the negative contribution in the focal tract**.
Label correctness	✗	The **Bespoke evidence chain is incomplete**. → **Discriminative** (after removing the unsupported claim)

Example from the Chicago ride-hailing case.

Benchmark and web platform

509 questions across 26 research cases

444 automatically generated · 65 human-written | Thesis-stage version



Generation · Review · Evaluation runs · Answer comparison

Agent error: reversed group labels

The calculation succeeded, but the agent misread the result.

Agent error example

Agent backbone: Qwen3.6 35B-A3B

Question

What concentration-model response was reported once open-water cover exceeded about 5% of a watershed?

Correct option: C. Its contribution decreased predicted dissolved-silicon concentration.

Returned results

aggregate · mean(FNConc)

land_Water	Mean
≤ 0.05	4.500
	Mean decreased ↓
> 0.05	3.313

Agent reading

Values assigned to the wrong groups

Mean	land_Water
3.31	≤ 0.05
↓ Mean increased ✕	
4.50	> 0.05

Agent answer: A. Its contribution became increasingly positive. ✕

Source: quiz_sysA_jgu-qwen3.6-35b_global_river_dissolved_silicon_concentration.jsonl

Answer records reveal errors that successful tool execution alone cannot show.

Main contributions

01 Evaluation framework

Designed a framework that connects GeoXAI tasks with their evidence requirements.

02 Automated workflow

Built a workflow for benchmark construction, quality checks and evaluation, with a supporting web platform.

03 Agent evaluation

Quantified how much GeoXAI-Agent helps compared with the same LLM used alone, across three LLM backbones.

Overall finding: the agent improved answer accuracy over LLM-only baselines.